

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Department

Trial HSC – Mathematics 2 Unit

August 2016

General Instructions

- Reading time – 5 minutes.
- Working time – 180 minutes.
- Approved calculators may be used.
- Write using blue or black pen.
- A BOSTES reference sheet is provided at the back of this paper. You may tear it off.
- In Question 11-16, show relevant mathematical reasoning and/or calculations.
- Begin each question on a new page of the answer booklet.
- Marks shown are a guide and may need to be adjusted.
- Full marks may not be awarded for careless work or illegible writing.

NAME: _____

TEACHER: _____

Total marks – 100

SECTION 1

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes.

SECTION 2

90 marks

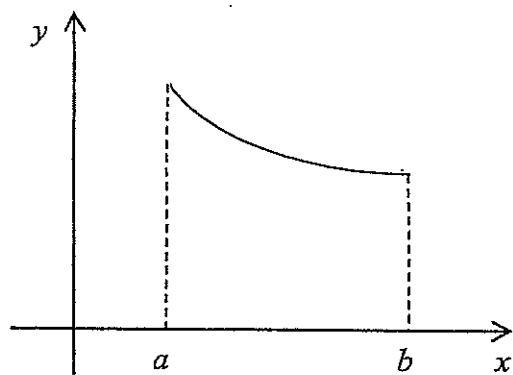
- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes.

Section 1**(10 marks)**

1. For what values of k does the equation $x^2 - 6x - 3k = 0$ have real roots?

- A) $k \geq -3$ B) $k \leq -3$ C) $k \geq 3$ D) $k \leq 3$

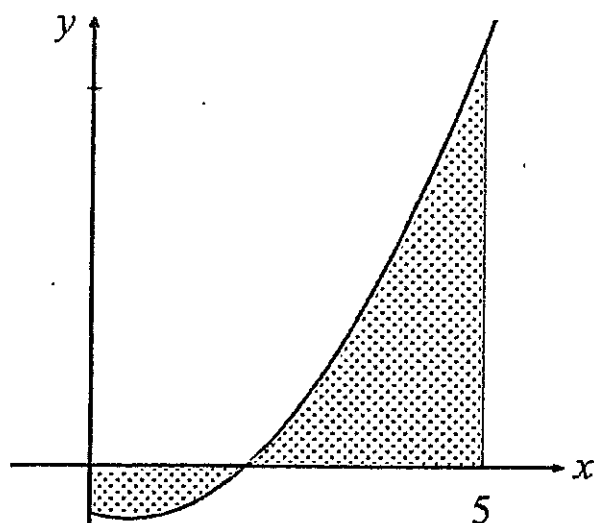
2. For the function $y = f(x)$, $a < x < b$ graphed below:



Which of the following is true?

- A) $f'(x) > 0$ and $f''(x) > 0$
B) $f'(x) > 0$ and $f''(x) < 0$
C) $f'(x) < 0$ and $f''(x) > 0$
D) $f'(x) < 0$ and $f''(x) < 0$

3. Which expression will give the area of the shaded region bounded by the curve $y = x^2 - x - 2$, the x -axis and the lines $x = 0$ and $x = 5$?



- A) $A = \left| \int_0^1 (x^2 - x - 2) dx \right| + \int_1^5 (x^2 - x - 2) dx$
- B) $A = \int_0^1 (x^2 - x - 2) dx + \left| \int_1^5 (x^2 - x - 2) dx \right|$
- C) $A = \left| \int_0^2 (x^2 - x - 2) dx \right| + \int_2^5 (x^2 - x - 2) dx$
- D) $A = \int_0^2 (x^2 - x - 2) dx + \left| \int_2^5 (x^2 - x - 2) dx \right|$
4. What are the coordinates of the focus of the parabola $4y = x^2 - 8$?
- A) (0, -8) B) (0, -7) C) (0, -2) D) (0, -1)

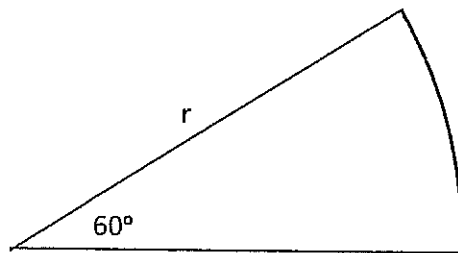
5. What are the domain and range of the function $f(x) = \sqrt{4 - x^2}$?

- A) Domain: $-2 \leq x \leq 2$, Range: $0 \leq y \leq 2$
B) Domain: $-2 \leq x \leq 2$, Range: $-2 \leq y \leq 2$
C) Domain: $0 \leq x \leq 2$, Range: $-4 \leq y \leq 4$
D) Domain: $0 \leq x \leq 2$, Range: $0 \leq y \leq 4$

6. When the curve $y = e^x$ is rotated about the x -axis between $x = -2$ and $x = 2$, the volume of the solid generated is given by:

- A) $\pi \int_{-2}^2 e^x dx$ B) $2\pi \int_0^2 e^{x^2} dx$
C) $\pi \int_{-2}^2 e^{x^2} dx$ D) $\pi \int_{-2}^2 e^{2x} dx$

7. The sector below has an area of 10π square units.



What is the value of r ?

- A) $\sqrt{60}$ B) $\pi\sqrt{60}$ C) $\sqrt{\frac{\pi}{3}}$ D) $\sqrt{\frac{1}{3}}$

8. An infinite geometric series has a first term of 8 and a limiting sum of 12. What is the common ratio?

A) $\frac{1}{6}$ B) $\frac{1}{4}$ C) $\frac{1}{3}$ D) $\frac{1}{2}$

9. If $\int_0^a 4 - 2x \, dx = 4$, find the value of a .

A) $a = -2$ B) $a = 0$ C) $a = 4$ D) $a = 2$

10. What is the greatest value taken by the function $f(x) = 4 - 2\cos x$ for $x \geq 0$?

A) 2 B) 4 C) 6 D) 8

Section 2**(90 marks)**

Question 11

(15 marks)

Marks

- a) Find $\sqrt[3]{9.8^2}$ correct to 2 decimal places 1
- b) Factorise fully $ax + 3ay - x - 3y$ 1
- c) Solve for a and d: 1

$$a + 9d = 20$$

$$2a + 9d = 12$$

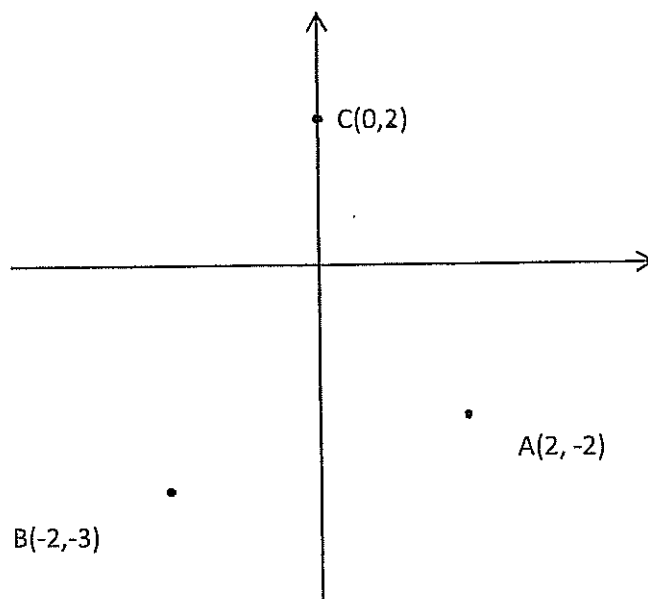
- d) Express $\frac{2}{5 + \sqrt{3}}$ with a rational denominator 1
- e) Solve $|3x - 1| = 5$ 2
- f) Solve the following equation: 2
- $$\log_2 x + \log_2(x + 7) = 3$$
- g) Solve $\cos x = \frac{-1}{2}$ for $0 \leq x \leq 2\pi$ 2
- h) Find the primitive of $x^2 \sqrt{x}$ 2
- i) Differentiate $\frac{3}{(2x+1)^2}$ 2
- j) Find $\int_0^1 e^{2x} dx$ 1

Question 12

(15 marks)

Marks

- a) On the diagram below, $A(2, -2)$, $B(-2, -3)$ and $C(0, 2)$ are the vertices of a triangle ABC. Copy this diagram into your answer booklet.



- | | | |
|------|--|---|
| i) | Find the gradient of AC | 1 |
| ii) | Find the angle of inclination that AC makes with the positive direction of the x axis, to the nearest degree. | 1 |
| iii) | Show that the equation of AC is $2x + y - 2 = 0$ | 1 |
| iv) | Calculate the perpendicular distance of B from the line AC | 2 |
| v) | Find the area of $\triangle ABC$ | 2 |
| vi) | Find the coordinates of D such that ABCD is a parallelogram. | 1 |
| b) | Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x}$ | 2 |
| c) | In $\triangle ABC$, $AB = 2\text{cm}$, $\angle ABC = 105^\circ$ and $\angle BCA = 30^\circ$. Find the length of BC correct to 1 d.p. | 2 |
| d) | Max is saving to buy a new car. He needs \$12700. In the first month he saves \$25, in the second \$40 followed by \$55 in the next. If he continues to increase the amount he saves by \$15 each month, how many months will it take him to save for the car? | 3 |

Question 13 (15 marks) Marks

a) Differentiate:

i) $x \tan 2x$ 2

ii) $e^{\sin x} + \frac{1}{x}$ 2

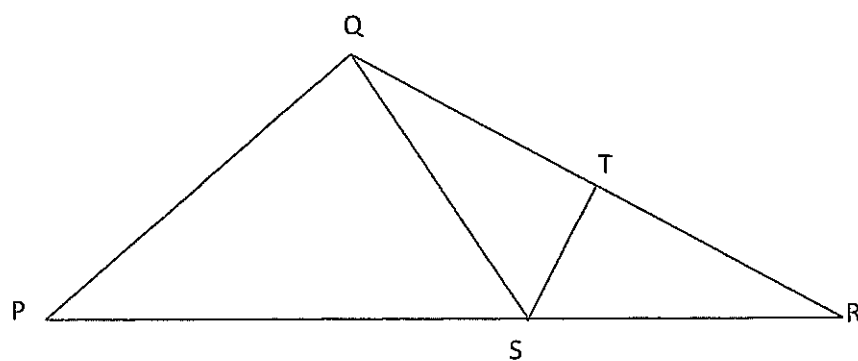
iii) $\frac{3x-7}{3+2x}$ 2

b) Find

i) $\int (5x - 1)^9 dx$ 2

ii) $\int \sin \frac{3x}{4} dx$ 2

c)



In $\triangle PQR$, point T lies on side QR and point S lies on side PR such that $QT = TR$,

$QS = QP$ and $ST \perp QT$.

i) Copy the diagram into your answer booklet showing all given information. 1

ii) Prove that $\triangle QTS \equiv \triangle RTS$ 2

iii) Prove that $\angle QPS = 2 \angle TQS$ 2

Question 14

(15 marks)

Marks

- a) Consider the curve

$$f(x) = -\frac{1}{3}x^3 - x^2 + 3x + 1$$

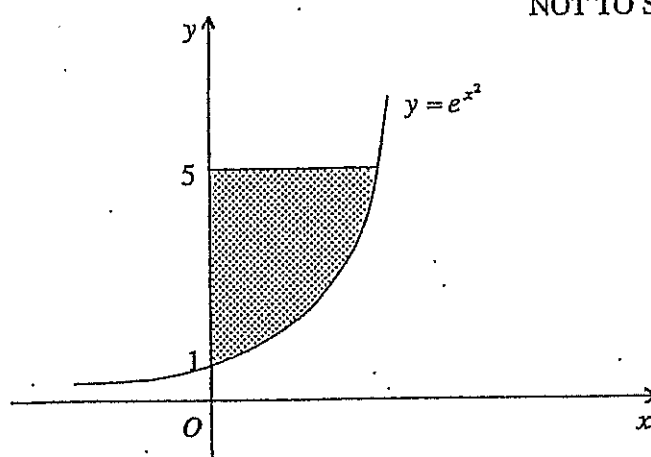
- i) Find the coordinates of any stationary points and determine their nature. 3
- ii) Find any point(s) of inflexion 2
- iii) Sketch the curve in the domain, $-6 \leq x \leq 3$ 2
- iv) What is the maximum value of $f(x)$ in the given domain? 1

- b) Simplify
- $\frac{1-\sin^2 x}{\cot x}$

2

- c)

NOT TO SCALE



The shaded region bounded by the graph $y = e^{x^2}$, the line $y = 5$ and the y axis is rotated about the y - axis to form a solid revolution.

- i) Show that the volume of the solid is given by 1

$$V = \pi \int_1^5 \log_e y \, dy$$

Marks

- ii) Copy and complete the following table into your writing booklet.

1

Give your answer correct to 3 decimal places.

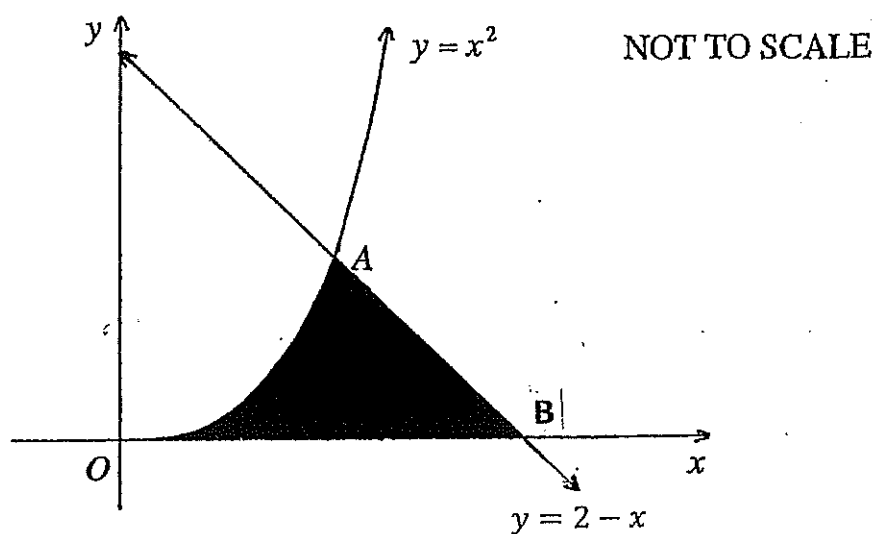
y	1	2	3	4	5
$\log_e y$	0	0.693	1.099		1.609

- iii) Use Simpson's Rule with five function values to approximate the volume

3

of the solid of revolution V_y , correct to three decimal places.

a)



The shaded region OAB is bounded by the parabola $y = x^2$, the line $y = 2 - x$ and the x -axis.

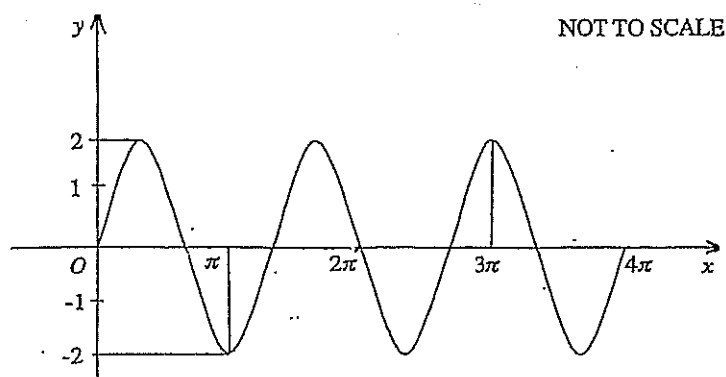
i) Find the x coordinates of A and B. 2

ii) Show that the exact area of the shaded region OAB is given by $\frac{5}{6}$ square units. 2

b) i) Show that $\frac{d}{dx}(xe^x) = e^x + xe^x$ 1

ii) Find $\int xe^x dx$ 2

c) Find the trigonometric equation for the graph below: 2



Question 15 (cont)

Marks

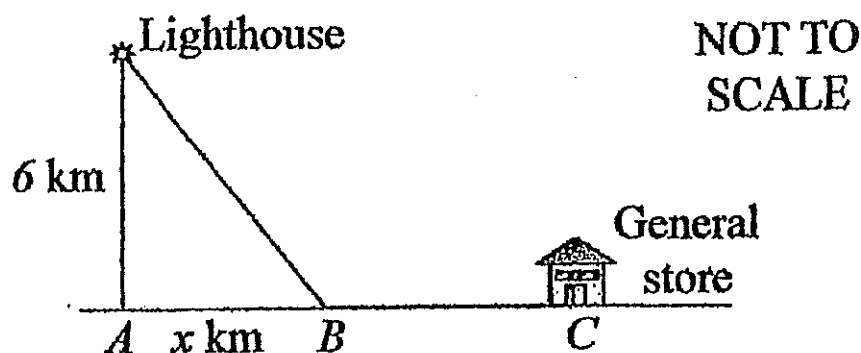
- d) Mr Egan borrows \$P from a bank to fund his house extensions. The term of the loan is 20 years with an annual interest rate of 9%. At the end of each month, interest is calculated on the balance owing and added to the balance owing. Mr Egan repays the loan in equal monthly instalments of \$1050.
- i) Write an expression for the amount, A_1 , Mr Egan owes at the end of the first month 1
- ii) Show that at the end of n months, the amount owing, A_n , is given by: 3
- $$A_n = P(1.0075)^n - 140000(1.0075)^n + 140000$$
- iii) If the loan is repaid at the end of 20 years, calculate the amount Mr Egan originally borrowed, correct to the nearest dollar. 2

Question 16

(15 marks)

- a) Find $\int 2^x dx$ 1
- b) Let α and β be the solutions of $x^2 + 5x + 3 = 0$. Find:
- i) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2
- ii) A quadratic equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ 2
- c) Evaluate $\int_0^2 \frac{6x}{x^2+2} dx$ 3

d)



The water's edge is a straight line ABC which runs east-west. A lighthouse is 6km from the shore on a rocky outcrop, due north of A.

10km due east of A is a general store. To get to the general store as quickly as possible the lighthouse keeper rows to a point B, x km from A, and then jogs to the general store.

The lighthouse keeper's rowing speed is 6km/h and his jogging speed is 10km/h.

- i) Show that it takes the lighthouse keeper $\frac{\sqrt{36+x^2}}{6}$ hours to row from the lighthouse to B. 2
- ii) Show that the total time taken for the lighthouse keeper to reach the general store is given by 1

$$T = \frac{\sqrt{36+x^2}}{6} + \frac{10-x}{10} \text{ hours}$$

- iii) Hence, show that when $x = 4\frac{1}{2}$ km, the time it takes the lighthouse keeper to travel from the lighthouse to the general store is a minimum (you may assume it is a minimum – no testing required) 3
- iv) Find the quickest time it takes the lighthouse keeper to go to the general store from the lighthouse. (You may leave your answer in hours). 1

Student Name: _____

Teacher Name: _____

2016 2 Unit Trial Solutions

Section 1

1. A 2. C 3. C 4. D 5. A 6. D 7. A
8. C 9. D 10. C

Section 2

Question 11

a) 4.58 b) $a(x+3y) - (x+3y)$ c) $a = -8$
 $(a-1)(x+3y)$ $d = \frac{28}{9}$

d) $\frac{2}{5+\sqrt{3}} \times \frac{5-\sqrt{3}}{5-\sqrt{3}}$ e) $3x-1=5$ $3x-1=-5$
 $= \frac{10-2\sqrt{3}}{22}$ $x=2$ $x = -\frac{4}{3}$
 $= \frac{5-\sqrt{3}}{11}$ ① ①

f) $\log_2 x(x+7) = 3$ g) $\cos x = -\frac{1}{2}$
 $x(x+7) = 8$ Working angle $\frac{\pi}{3}$
 $x^2 + 7x - 8 = 0$ ✓ $\begin{array}{c|c} 5 & A \\ \hline 7 & C \end{array}$
 $(x-1)(x+8) = 0$
 $x=1$ or $x=-8$
 $x=1$ or $x > 0$ ① ①
 $\therefore x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$
 $x = \frac{2\pi}{3}, \frac{4\pi}{3}$ ①

h) $x^2 \sqrt{x}$ i) $\frac{d}{dx} (2x+1)^2$ j) $\int_0^1 e^{2x} dx$
 $\int x^{\frac{5}{2}} dx$ ① $\frac{d}{dx} 3x(2x+1)^{-2}$ $\left[\frac{1}{2} e^{2x} \right]_0^1$
 $= \frac{2}{7} x^{\frac{7}{2}} + C$ $= -12(2x+1)^{-3}$ ① $\frac{1}{2}(e^2 - 1)$

Student Name: _____

Teacher Name: _____

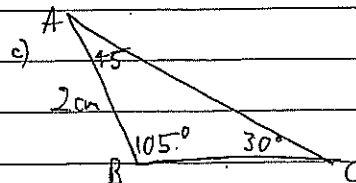
Question 12

a) $m_{AC} = \frac{2-2}{0-2} = \frac{0}{-2} = 0$ c) $m = \tan \theta$ d) $y-2 = -2(x-0)$
 $-2 = \tan \theta$ $y-2 = -2x$
 $m_{AC} = -2$ $\theta = 117^\circ$ $2x + y - 2 = 0$

c) $d = \frac{|2x-2+1x-3+-2|}{\sqrt{2^2+1^2}}$ d) $A = \frac{1}{2} \times AC \times \frac{9}{\sqrt{5}}$
 $= \frac{9}{\sqrt{5}}$ units ① $= \frac{9}{\sqrt{5}} \times \sqrt{2^2+(-2)^2}$
 $= \frac{9}{\sqrt{5}} \times \sqrt{20}$ ①
 $= \frac{9}{\sqrt{5}} \times 2\sqrt{5}$
 $= 9 \text{ units}^2$ ①

vi) D(4,3)

b) $\frac{2}{3} \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$ ①
 $= \frac{2}{3}$ ①



$\frac{BC}{\sin 45} = \frac{2}{\sin 30}$ ①
 $BC = \frac{2 \sin 45}{\sin 30}$
 $BC = 2.8 \text{ cm}$ ①

d) $12700 = 25 + 40 + 55 + \dots$ ①
 $S_n = 12700$ a, $d = 15$, $n = ?$ ①
 $12700 = \frac{n}{2} (2 \times 25 + (n-1)15)$ $15n^2 + 35n - 25400 = 0$
 $25400 = 50n + 15n^2 - 15n$ $3n^2 + 7n - 5080 = 0$
 $n = \frac{-7 \pm \sqrt{49 + 12 \times 5080}}{6}$ ①
 $n = 40$ ($n > 0$)

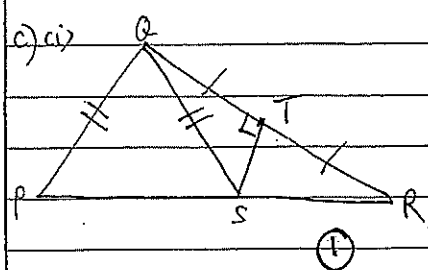
Question 13

a) i) $\frac{d}{dx}(x \tan 2x)$
 $= \tan 2x + x \times 2 \sec^2 2x$ ①
 $= \tan 2x + 2x \sec^2 2x$ ①

ii) $\frac{d}{dx}(e^{\sin x} + x^{-1})$
 $= \cos x e^{\sin x} - x^{-2}$
 ① ①

c) i) $\frac{d}{dx} \left(\frac{3x-7}{3+2x} \right)$
 $\frac{(3+2x)3 - (3x-7)2}{(3+2x)^2}$ ①
 $= \frac{23}{(3+2x)^2}$ ①

b) i) $\int (5x-1)^9 dx$ ii) $\int \sin \frac{3x}{4} dx$
 $= \frac{(5x-1)^{10}}{50} + C$ ②
 $= -\frac{4}{3} \cos \frac{3x}{4} + C$ ②

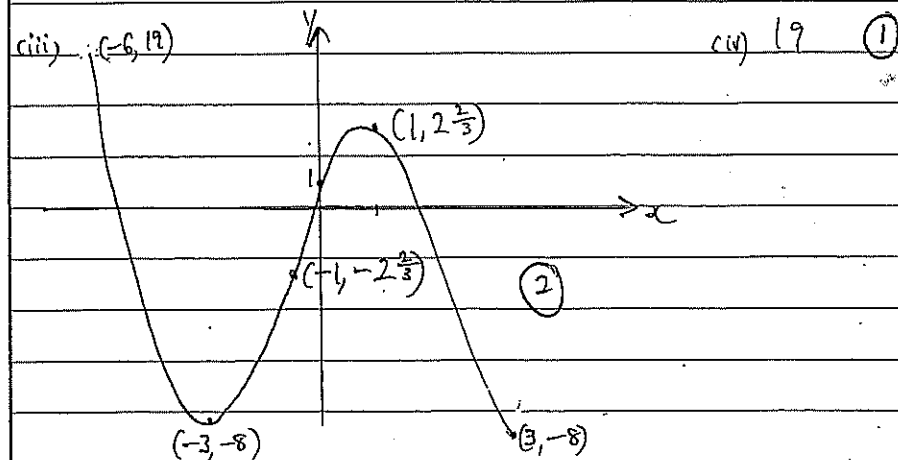


c) ii) In Δ 's QTS and RTS,
 ST is common
 QT = TR given
 $\angle QTS = \angle RTS$ (straight angle ST $\perp$ QR)
 $\therefore \Delta QTS \equiv \Delta RTS$ (SAS) ①

c) iii) Let $\angle TQS = \theta$
 $\therefore \angle TRS = \theta$ (corresponding angles in congruent Δ 's)
 $\therefore \angle QST = \angle RST = 90 - \theta$ (angle sum of Δ 's)
 $\angle QSP = 180 - 2(90 - \theta)$ (straight angle)
 $= 2\theta$
 $\angle QPS = 2\theta$ (equal angles opposite equal sides of a triangle)
 $\therefore \angle QPS = 2 \angle TQS$ ①

Question 14

a) $f(x) = \frac{1}{3}x^3 - x^2 + 3x + 1$ ii) $f''(x) = 0$ for pts. of inflexion
 $f'(x) = -x^2 - 2x + 3 = 0$
 $x^2 + 2x - 3 = 0$ ①
 $(x-1)(x+3) = 0$
 $x = 1$ or -3
 $f''(x) = -2x - 2$ ①
 $f''(1) = -4 < 0 \therefore (1, 2\frac{2}{3})$ max. horizontal inflexion
 $f''(-3) = 4 > 0 \therefore (-3, -8)$ min. as $f'(-1) \neq 0$ No testing required.
 ①



b) $\frac{\cos^2 x}{\cos x \sin x}$ ①
 $= \frac{\cos^2 x}{1} \times \frac{\sin x}{\cos x}$
 $= \sin x \cos x$ ①

c) i) $V = \pi \int x^2 dy$
 If $y = e^{x^2}$ then $\log_e y = x^2$
 $\therefore V = \pi \int_1^5 \log_e y dy$ ①

Student Name: _____

Teacher Name: _____

ii)	y	1	2	3	4	5
	$\log_e y$	0	0.693	1.099	1.386	1.609

①

$$\text{iii) } V_y \div \frac{1}{3} \{ 0 + 1.609 + 4(0.693 + 1.386) + 2 \times 1.099 \} \times \pi$$

$$= 12.695 \quad \text{②}$$

Student Name: _____

Teacher Name: _____

Question 15

$$\text{ai) } x^2 = 2 - x$$

$$y = 2 - x$$

$$x^2 + x - 2 = 0$$

$$0 = 2 - x$$

$$(x-1)(x+2) = 0$$

$$\therefore x = 2 \text{ at } B \quad \text{①}$$

$$x = 1 \text{ or } -2$$

$$\text{At } A \quad x = 1, (>0) \quad \text{①}$$

$$\text{ai) } A = \int_0^1 x^2 dx + \int_1^2 2-x dx$$

$$= \left[\frac{x^3}{3} \right]_0^1 + \left[2x - \frac{x^2}{2} \right]_1^2 \quad \text{①}$$

$$= \frac{1}{3} + (4-2) - (2-\frac{1}{2})$$

$$= \frac{1}{3} + 2 - 2 + \frac{1}{2}$$

$$= \frac{5}{6} \text{ units}^2 \quad \text{①}$$

$$\text{b) ai) } \frac{d}{dx}(xe^x)$$

$$= xe^x + e^x \times 1$$

$$= e^x + xe^x \quad \text{①}$$

$$\text{ii) } \frac{d}{dx}(xe^x) = e^x + xe^x$$

$$\frac{d}{dx}(xe^x) - e^x = xe^x \quad \text{①}$$

$$\int \frac{d}{dx}(xe^x) - e^x dx = \int xe^x dx$$

$$xe^x - e^x + C = \int xe^x dx \quad \text{①}$$

$$\text{c) Amplitude } 2$$

$$\text{Period} = \frac{4\pi}{3}$$

$$\therefore T = \frac{2\pi}{n} = \frac{4\pi}{3} \therefore n = \frac{3}{2}$$

Curve is of the

$$\text{form } y = A \sin nx$$

$$\therefore y = 2 \sin \frac{3x}{2}$$

①

①

$$d) \text{ci) } A_1 = P_x \left(1 + \frac{7.5}{100}\right) - 1050$$

$$A_1 = P_x (1.0075) - 1050 \quad (1)$$

$$\text{cii) } A_2 = A_1 \times 1.0075 - 1050$$

$$= [P_x \times 1.0075 - 1050] \times 1.0075 - 1050$$

$$= P_x \times 1.0075^2 - 1050(1 + 1.0075)$$

$$A_n = P_x \times 1.0075^n - 1050(1 + 1.0075 + \dots + 1.0075^{n-1}) \quad (1)$$

$a=1, r=1.0075, n=n$

$$= P_x \times 1.0075^n - 1050 \times \frac{1 \times 1.0075^n - 1}{1.0075 - 1} \quad (1)$$

$$= P_x \times 1.0075^n - 140000(1.0075^n - 1)$$

$$= P_x \times 1.0075^n - 140000 \times 1.0075^n + 140000 \quad (1)$$

ci) At 20 years, $n=240$, $A_n=0$, solve P

$$0 = P_x 1.0075^{240} - 140000 \times 1.0075^{240} + 140000$$

$$P = \$116702 \quad (2)$$

Question 16

$$a) \int 2^x dx$$

$$\frac{1}{\log_e 2} 2^x + C \quad (1)$$

$$b) \text{ci) } \alpha + \beta = -5$$

$$\alpha \beta = 3$$

$$\frac{\alpha + \beta}{\alpha \beta} = \frac{-5}{3} \quad (1)$$

$$= \frac{-5}{3} \quad (1)$$

$$\text{cii) } x^2 - \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)x + \frac{1}{\alpha\beta} = 0$$

$$x^2 - \left(-\frac{5}{3}\right)x + \frac{1}{3} = 0$$

$$3x^2 + 5x + 1 = 0$$

$$x^2 + \frac{5}{3}x + \frac{1}{3} = 0$$

$$c) \int_0^2 \frac{6x}{x^2+2} dx$$

$$3 \int_0^2 \frac{2x}{x^2+2} dx \quad (1)$$

$$= 3 \left[\log_e(x^2+2) \right]_0^2 \quad (1)$$

$$= 3 [\log_e 6 - \log_e 2]$$

$$= 3 \log_e 3 \quad (1)$$

$$d) \text{ci) } \triangle \frac{D}{5 \times T} \quad T = \frac{D}{S}$$

distance from B to light house

$$= \sqrt{x^2 + 36} \text{ km} \quad (1)$$

$$\therefore T = \frac{\sqrt{x^2 + 36}}{6} \text{ hours} \quad (1)$$

cii) Running:

$$T = \frac{\text{distance BC}}{\text{running speed}}$$

$$= \frac{10-x}{10}$$

$\therefore$ Total time T

$$= \frac{\sqrt{x^2 + 36}}{6} + \frac{10-x}{10}$$

$$\text{ciii) } \frac{dT}{dx} = \frac{1}{6} \times \frac{1}{2}(x^2 + 36)^{-\frac{1}{2}} \times 2x - \frac{1}{10}$$

$$= \frac{x}{6\sqrt{x^2 + 36}} - \frac{1}{10} = 0 \text{ for a}$$

(1)

minimum.

Student Name: _____

Teacher Name: _____

$$\frac{x}{6\sqrt{x^2+36}} = \frac{1}{10}$$

$$10x = 6\sqrt{x^2+36}$$

$$\frac{5x}{3} = \sqrt{x^2+36}$$

$$\frac{25x^2}{9} = x^2 + 36 \quad (1)$$

$$25x^2 = 9x^2 + 324$$

$$16x^2 = 324$$

$$x^2 = 20.25$$

$$x = 4.5 \text{ Km} \quad (1)$$

civ) Sub $x = 4.5$ into expression for T .

$$T = \frac{\sqrt{4.5^2 + 36}}{6} + \frac{10 - 4.5}{10}$$

$$= 1.8 \text{ hours} \quad (1)$$